Unit : 2 PROGRAMS

- Write a servlet to print Hello World.

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/hello")
public class HelloServlet extends HttpServlet {

   protected void doGet(HttpServletRequest request,
                        HttpServletResponse response)
                        throws ServletException, IOException {

      response.setContentType("text/html");

      PrintWriter out = response.getWriter();

      out.println("<html>");
      out.println("<body>");
      out.println("<h1>Hello World</h1>");
      out.println("</body>");
      out.println("</html>");
   }
}
```

- Write a servlet that displays today's date and time.

```
import java.io.IOException;
import java.io.PrintWriter;
import java.util.Date;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
```

```
@WebServlet("/datetime")
public class DateTimeServlet extends HttpServlet {

   protected void doGet(HttpServletRequest request,
                        HttpServletResponse response)
                        throws ServletException, IOException {

      response.setContentType("text/html");

      PrintWriter out = response.getWriter();

      Date d = new Date();

      out.println("<html>");
      out.println("<body>");

      out.println("<h1>Today's Date and Time</h1>");
      out.println("<h2>" + d + "</h2>");

      out.println("</body>");
      out.println("</html>");
   }
}
```

- Write a program that accepts username and after submitting the form, welcomes the user by writing "Welcome Username".

Create **`index.html`**
In NetBeans, go to **Web Pages → New → HTML** and create:

```
<!DOCTYPE html>
<html>
<head>
   <title>Welcome User</title>
</head>
<body>

   <h2>Enter Your Name</h2>

   <form action="welcome" method="get">

      Enter Username:
      <input type="text" name="username">

      <input type="submit" value="Submit">

   </form>
```

```
</body>
</html>
```

Create WelcomeServlet.java
Right-click **Source Packages → New → Servlet**.

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/welcome")
public class WelcomeServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");

        out.println("<html>");
        out.println("<body>");

        out.println("<h1>Welcome " + username + "</h1>");

        out.println("</body>");
        out.println("</html>");
    }
}
```

- Write a program that accepts Employee Number, Employee Name, Designation, Qualification and after submitting the form display that information on a page using GET method.

employee.html

```
<!DOCTYPE html>
<html>
<head>
```

```
    <title>Employee Form</title>
</head>
<body>

    <h2>Employee Information</h2>

    <form action="employee" method="get">

        Employee Number:
        <input type="text" name="empno">
        <br><br>

        Employee Name:
        <input type="text" name="empname">
        <br><br>

        Designation:
        <input type="text" name="designation">
        <br><br>

        Qualification:
        <input type="text" name="qualification">
        <br><br>

        <input type="submit" value="Submit">

    </form>

</body>
</html>
```

EmployeeServlet.java page

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/employee")
```

```
public class EmployeeServlet extends HttpServlet {

   protected void doGet(HttpServletRequest request,
                        HttpServletResponse response)
                        throws ServletException, IOException {

      response.setContentType("text/html");

      PrintWriter out = response.getWriter();

      // Get values from HTML form
      String empno = request.getParameter("empno");
      String empname = request.getParameter("empname");
      String designation = request.getParameter("designation");
      String qualification = request.getParameter("qualification");

      // Display values
      out.println("<html>");
      out.println("<body>");

      out.println("<h2>Employee Information</h2>");

      out.println("Employee Number : " + empno + "<br><br>");
      out.println("Employee Name : " + empname + "<br><br>");
      out.println("Designation : " + designation + "<br><br>");
      out.println("Qualification : " + qualification + "<br><br>");

      out.println("</body>");
      out.println("</html>");
   }
}
```

- Write a program that register a student by providing rollno, name, course, semester, hobbies, and after submitting the form display the information on a page using POST method. (Hint : By using getParameterNames() and getParameterValues())

student.html

```
<!DOCTYPE html>
<html>
<head>
```

```
    <title>Student Registration</title>
</head>
<body>

    <h2>Student Registration Form</h2>

    <form action="student" method="post">

        Roll No:
        <input type="text" name="rollno">
        <br><br>

        Name:
        <input type="text" name="name">
        <br><br>

        Course:
        <input type="text" name="course">
        <br><br>

        Semester:
        <input type="text" name="semester">
        <br><br>

        Hobbies:
        <input type="checkbox" name="hobby" value="Reading"> Reading

        <input type="checkbox" name="hobby" value="Music"> Music

        <input type="checkbox" name="hobby" value="Cricket"> Cricket

        <input type="checkbox" name="hobby" value="Travelling"> Travelling

        <br><br>

        <input type="submit" value="Register">

    </form>

</body>
</html>
```

StudentServlet.java Page

```
import java.io.IOException;
import java.io.PrintWriter;
import java.util.Enumeration;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/student")
public class StudentServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<html>");
        out.println("<body>");

        out.println("<h2>Student Registration Information</h2>");

        // Get parameter names
        Enumeration<String> names = request.getParameterNames();

        while (names.hasMoreElements()) {

            String name = names.nextElement();

            // Get multiple values
            String values[] = request.getParameterValues(name);

            out.println("<b>" + name + " : </b>");

            for (String value : values) {
                out.println(value + " ");
            }
```

```
            out.println("<br><br>");
        }

        out.println("</body>");
        out.println("</html>");
    }
}
```

- Demonstrate the Servlet lifecycle with appropriate messages in each stage.

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/lifecycle")
public class LifeCycleServlet extends HttpServlet {

    // Called only once when servlet is created
    @Override
    public void init() throws ServletException {
        System.out.println("1. init() method called");
    }

    // Called for every request
    @Override
    protected void service(HttpServletRequest request,
                           HttpServletResponse response)
                           throws ServletException, IOException {

        System.out.println("2. service() method called");

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<html>");
        out.println("<body>");

        out.println("<h2>Servlet Life Cycle</h2>");
```

```
        out.println("<p>init() method called</p>");
        out.println("<p>service() method called</p>");
        out.println("<p>destroy() method will be called when servlet is stopped</p>");

        out.println("</body>");
        out.println("</html>");
    }

    // Called only once when servlet is destroyed
    @Override
    public void destroy() {
        System.out.println("3. destroy() method called");
    }
}
```

- Write a servlet program that displays the Basic Header Information.

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/header")
public class HeaderServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<html>");
        out.println("<body>");

        out.println("<h2>Basic Header Information</h2>");

        out.println("Request Method : " + request.getMethod() + "<br><br>");
```

```
        out.println("Request URI : " + request.getRequestURI() + "<br><br>");

        out.println("Protocol : " + request.getProtocol() + "<br><br>");

        out.println("Server Name : " + request.getServerName() + "<br><br>");

        out.println("Server Port : " + request.getServerPort() + "<br><br>");

        out.println("Remote Address : " + request.getRemoteAddr() + "<br><br>");

        out.println("User Agent : " + request.getHeader("User-Agent") + "<br><br>");

        out.println("Host : " + request.getHeader("Host") + "<br><br>");

        out.println("</body>");
        out.println("</html>");
    }
}
```

- Write a program that creates a login form and after clicking on login button if login details are correct than displays welcome screen else display appropriate error message.(Hint : take username="admin", password = "admin")

login.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Login Form</title>
</head>
<body>

    <h2>Login Form</h2>

    <form action="login" method="post">

        Username:
        <input type="text" name="username">
        <br><br>

        Password:
        <input type="password" name="password">
```

```
        <br><br>

        <input type="submit" value="Login">

    </form>

</body>
</html>
```

LoginServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        // Get username and password
        String username = request.getParameter("username");
        String password = request.getParameter("password");

        // Check login details
        if (username.equals("admin") && password.equals("admin")) {

            out.println("<html>");
            out.println("<body>");

            out.println("<h1>Welcome Admin</h1>");
            out.println("<h2>Login Successful</h2>");
```

```
            out.println("</body>");
            out.println("</html>");

        } else {

            out.println("<html>");
            out.println("<body>");

            out.println("<h2>Login Failed</h2>");
            out.println("<p>Invalid Username or Password</p>");

            out.println("</body>");
            out.println("</html>");
        }
    }
}
```

- Write a Servlet Program that gives the following option using the radio button and perform appropriate operation (Addition , Subtraction, Multiplication, Division).

  .html file

```
<html>
<head> </head>
<body>
<form action="calculator" method="post">

    Enter First Number:
    <input type="text" name="num1">
    <br><br>

    Enter Second Number:
    <input type="text" name="num2">
    <br><br>

    <input type="radio" name="operation" value="add"> Addition
    <br>

    <input type="radio" name="operation" value="sub"> Subtraction
    <br>

    <input type="radio" name="operation" value="mul"> Multiplication
```

```
    <br>

    <input type="radio" name="operation" value="div"> Division
    <br><br>

    <input type="submit" value="Calculate">

</form>

</body>
</html>
```

Servlet file

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/calculator")
public class CalculatorServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        double num1 = Double.parseDouble(request.getParameter("num1"));
        double num2 = Double.parseDouble(request.getParameter("num2"));

        String operation = request.getParameter("operation");

        double result = 0;

        if(operation.equals("add"))
            result = num1 + num2;
```

```
        else if(operation.equals("sub"))
            result = num1 - num2;

        else if(operation.equals("mul"))
            result = num1 * num2;

        else if(operation.equals("div"))
            result = num1 / num2;

        out.println("<html>");
        out.println("<body>");

        out.println("<h2>Calculator Result</h2>");
        out.println("<h3>Answer = " + result + "</h3>");

        out.println("</body>");
        out.println("</html>");
    }
}
```

- Write a Servlet Program that redirects a page to google.com & youtube.com.

.html file

```
<html>
<head></head>
<body>

  <form action="redirect" method="get">

    <input type="submit" name="site" value="Google">

    <input type="submit" name="site" value="YouTube">

  </form>

</body>
</html>
```

Servlet file

```
import java.io.IOException;

import jakarta.servlet.ServletException;
```

```
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/redirect")
public class RedirectServlet extends HttpServlet {

   @Override
   protected void doGet(HttpServletRequest request,
                        HttpServletResponse response)
                        throws ServletException, IOException {

      String site = request.getParameter("site");

      if(site.equals("Google"))
      {
         response.sendRedirect("https://www.google.com");
      }
      else if(site.equals("YouTube"))
      {
         response.sendRedirect("https://www.youtube.com");
      }
   }
}
```

- Write a Servlet Program that sends a customized error message for “404 – Page Not Found”.

  Index.html

```
<html>
<head></head>
<body>
   <form action="ErrorServlet" method="get">

      Enter Page Name:
      <input type="text" name="page">

      <input type="submit" value="Open">

   </form>
</body>
</html>
```

home.html

```
<html>
<head>
   <title>Home</title>
</head>
<body>
   <h1>Welcome to Home Page</h1>
</body>
</html>
```

Servlet file

```
import java.io.IOException;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/ErrorServlet")
public class ErrorServlet extends HttpServlet {

   @Override
   protected void doGet(HttpServletRequest request,
                  HttpServletResponse response)
                  throws ServletException, IOException {

      String page = request.getParameter("page");

      if(page.equals("home"))
      {
         response.sendRedirect("home.html");
      }
      else
      {
         response.sendError(404,
               "Page Not Found. The page '" + page + "' does not exist.");
      }
   }
}
```

- Write a Servlet program that demonstrates the use of URL Rewriting.

Index.html

```
<html>

<head></head>
<body>
   <h2>Enter Username</h2>

   <form action="FirstServlet" method="get">

      Username :
      <input type="text" name="username">

      <br><br>

      <input type="submit" value="Next">

   </form>

</body>
</html>
```

FirstServlet file

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/FirstServlet")
public class FirstServlet extends HttpServlet {

   @Override
   protected void doGet(HttpServletRequest request,
                        HttpServletResponse response)
                        throws ServletException, IOException {

      response.setContentType("text/html");

      PrintWriter out = response.getWriter();

      String username = request.getParameter("username");
```

```
        out.println("<html>");
        out.println("<body>");

        out.println("<h2>First Servlet</h2>");

        out.println("<a href='SecondServlet?username=" + username + "'>");
        out.println("Click Here to Go to Second Servlet");
        out.println("</a>");

        out.println("</body>");
        out.println("</html>");
    }
}
```

SecondServlet file

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/SecondServlet")
public class SecondServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");

        out.println("<html>");
        out.println("<body>");

        out.println("<h2>Second Servlet</h2>");

        out.println("<h3>Welcome " + username + "</h3>");

        out.println("</body>");
        out.println("</html>");
```

```
    }
}
```

- Write a Servlet program that demonstrates the use of Hidden Form Field.

.html file

```
<html>
<head></head>
<body>

<h2>Enter Username</h2>

<form action="FirstServlet" method="post">

    Username :
    <input type="text" name="username">
    <br><br>
    <input type="submit" value="Next">

</form>

</body>
</html>
```

FirstServlet file

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/FirstServlet")
public class FirstServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();
```

```
        String username = request.getParameter("username");

        out.println("<html>");
        out.println("<body>");

        out.println("<h2>First Servlet</h2>");

        out.println("<form action='SecondServlet' method='post'>");

        // Hidden Form Field
        out.println("<input type='hidden' name='username' value='" + username + "'>");

        out.println("<input type='submit' value='Go To Second Servlet'>");

        out.println("</form>");

        out.println("</body>");
        out.println("</html>");
    }
}
```

SecondServlet file

```
package com.mycompany.demoprogramsservlet;

import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/SecondServlet")
public class SecondServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");
```

```
        out.println("<html>");
        out.println("<body>");

        out.println("<h2>Second Servlet</h2>");
        out.println("<h3>Welcome " + username + "</h3>");

        out.println("</body>");
        out.println("</html>");
    }
}
```

- Write a servlet program that checks if a cookie exists; if not, it creates a new cookies and adds it to the response. The servlet should then display all the cookies sent by the browser.

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/CookieServlet")
public class CookieServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        Cookie cookies[] = request.getCookies();

        out.println("<html>");
        out.println("<body>");
```

```
        // Check if cookies exist
        if (cookies == null) {

            Cookie c = new Cookie("username", "Hello");

            response.addCookie(c);

            out.println("<h2>New Cookie Created Successfully.</h2>");
            out.println("<p>Please Refresh the Page.</p>");

        } else {
            out.println("<h2>Cookies Received From Browser</h2>");
            for (Cookie c : cookies) {

                out.println("<b>Name :</b> " + c.getName() + "<br>");
                out.println("<b>Value :</b> " + c.getValue() + "<br><br>");
            }
        }
        out.println("</body>");
        out.println("</html>");
    }
}
```

- Write a servlet that displays a "Welcome" message if the user is visiting the site for the first time, and "Welcome back" if they have visited before.(Using Cookie)

Servlet file

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/WelcomeCookieServlet")
public class WelcomeCookieServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response) throws ServletException, IOException {
```

```
        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        Cookie cookies[] = request.getCookies();

        boolean found = false;

        if (cookies != null) {

           for (Cookie c : cookies) {

              if (c.getName().equals("username")) {

                 found = true;
                 break;
              }
           }
        }
        out.println("<html>");
        out.println("<body>");

        if (found) {
           out.println("<h2>Welcome Back!</h2>");
        } else {
           out.println("<h2>Welcome! You are visiting for the first time.</h2>");

           Cookie c = new Cookie("username", "visitor");

           response.addCookie(c);
        }
        out.println("</body>");
        out.println("</html>");
     }
  }
```

- Develop a servlet that allows the user to select a background color from a dropdown list . save the selected color in the cookie and apply it to the response page.

  Index.html

```
<html>
<head></head>
<body>

<h2>Select Background Color</h2>

<form action="ColorServlet" method="post">

   <select name="color">

      <option value="white">White</option>
      <option value="yellow">Yellow</option>
      <option value="lightblue">Light Blue</option>
      <option value="lightgreen">Light Green</option>
      <option value="pink">Pink</option>
   </select>
   <br><br>
   <input type="submit" value="Apply Color">
</form>

</body>
</html>
```

Servlet File

```
import java.io.IOException;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/ColorServlet")
public class ColorServlet extends HttpServlet {

   @Override
   protected void doPost(HttpServletRequest request,
                  HttpServletResponse response)
                  throws ServletException, IOException {

      String color = request.getParameter("color");
```

```
        Cookie c = new Cookie("bgcolor", color);

        response.addCookie(c);

        response.sendRedirect("DisplayServlet");
    }
}
```

Display Servlet File

```
package com.mycompany.demoprogramsservlet;

import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/DisplayServlet")
public class DisplayServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        String color = "white";

        Cookie cookies[] = request.getCookies();

        if (cookies != null) {

            for (Cookie c : cookies) {

                if (c.getName().equals("bgcolor")) {
```

```
                color = c.getValue();
                break;
            }
          }
        }

        out.println("<html>");
        out.println("<body bgcolor='" + color + "'>");

        out.println("<h2>Background Color Applied Successfully</h2>");
        out.println("<h3>Selected Color : " + color + "</h3>");

        out.println("</body>");
        out.println("</html>");
    }
}
```

- Write a servlet program that accepts a user's name through a form and stores it in an HTTP session. On subsequent visits during the session, greet the user by name.

  Index.html

```
<html>
<head></head>
<body>

<h2>Enter Your Name</h2>

<form action="SessionServlet" method="post">

   Name :
   <input type="text" name="username">

   <br><br>

   <input type="submit" value="Submit">
</form>
</body>
</html>
```

  Servlet File

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/SessionServlet")
public class SessionServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession();

        String name = request.getParameter("username");

        // First visit
        if (name != null && !name.equals("")) {

            session.setAttribute("username", name);

            out.println("<html>");
            out.println("<body>");

            out.println("<h2>Welcome " + name + "</h2>");
            out.println("<h3>Your name has been stored in the session.</h3>");

            out.println("<br>");
            out.println("<a href='SessionServlet'>Visit Again</a>");

            out.println("</body>");
            out.println("</html>");
        }
```

```
    }

    @Override
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession();

        String name = (String) session.getAttribute("username");

        out.println("<html>");
        out.println("<body>");

        if (name != null) {
            out.println("<h2>Welcome Back " + name + "</h2>");
        } else {
            out.println("<h2>No Session Found</h2>");
            out.println("<a href='index.html'>Go to Login Page</a>");
        }

        out.println("</body>");
        out.println("</html>");
    }
}
```

- Write a Java Servlet application that authenticates users using a login form. Upon successful login, store the user's information (like username) in the HTTP session. Display a personalized welcome page using session data . For the unsuccessful attempt display appropriate message. Also , implement a logout function that invalidates the session.

  Login.html

```
<html>
<head>
    <title>Login Form</title>
</head>
```

```
<body>

<h2>Login Form</h2>

<form action="LoginServlet" method="post">

   Username :
   <input type="text" name="username">

   <br><br>

   Password :
   <input type="password" name="password">

   <br><br>

   <input type="submit" value="Login">

</form>

</body>
</html>
```

LoginServlet

```
import java.io.IOException;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {

   @Override
   protected void doPost(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {

      String username = request.getParameter("username");
      String password = request.getParameter("password");
```

```
        if(username.equals("admin") && password.equals("admin"))
        {
            HttpSession session = request.getSession();

            session.setAttribute("username", username);

            response.sendRedirect("WelcomeServlet");
        }
        else
        {
            response.setContentType("text/html");

            response.getWriter().println("<h2>Invalid Username or Password</h2>");
            response.getWriter().println("<a href='login.html'>Try Again</a>");
        }
    }
}
```

Welcome Servlet

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/WelcomeServlet")
public class WelcomeServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession(false);
```

```
        if(session != null)
        {
            String username = (String)session.getAttribute("username");

            out.println("<html>");
            out.println("<body>");

            out.println("<h2>Welcome " + username + "</h2>");
            out.println("<h3>Login Successful</h3>");

            out.println("<br>");

            out.println("<a href='LogoutServlet'>Logout</a>");

            out.println("</body>");
            out.println("</html>");
        }
        else
        {
            out.println("<h2>Session Expired</h2>");
            out.println("<a href='login.html'>Login Again</a>");
        }
    }
}
```

Logout Servlet

```
import java.io.IOException;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/LogoutServlet")
public class LogoutServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {
```

```
        HttpSession session = request.getSession(false);

        if(session != null)
        {
           session.invalidate();
        }

        response.setContentType("text/html");

        response.getWriter().println("<h2>You have been logged out successfully.</h2>");
        response.getWriter().println("<a href='login.html'>Login Again</a>");
    }
}
```

- Create a servlet that counts how many times a user has visited the page during a session. Display the visit count on each load.

  Servlet file

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/VisitCounterServlet")
public class VisitCounterServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
                         throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession();
```

```
        Integer count = (Integer) session.getAttribute("count");

        if (count == null) {
           count = 1;
        } else {
           count++;
        }

        session.setAttribute("count", count);

        out.println("<html>");
        out.println("<body>");

        out.println("<h2>Session Visit Counter</h2>");

        out.println("<h3>You have visited this page <font color='blue'>"
                + count + "</font> times during this session.</h3>");

        out.println("<br>");
        out.println("<a href='VisitCounterServlet'>Visit Again</a>");

        out.println("</body>");
        out.println("</html>");
    }
}
```